

Lesson 09 · Marginal quantities and local estimates

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

If the demand function is $P = 100 - 4Q$, find an expression for marginal revenue in terms of Q .

☐ A $100 - 4Q$

☐ B $100 - 8Q$

☐ C $100Q - 4Q^2$

☐ D -4

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Q2 · Rank 1 / Foundation

If the average cost function is $AC = 15/Q + 2Q + 9$, what are the fixed costs?

☐ A 9

☐ B 0

☐ C 15

☐ D 2

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Q3 · Rank 1 / Foundation

A firm has production function $Q = 50L - 0.01L^2$, where L denotes the workforce. Find the marginal product of labour when $L = 10$.

☐ A 499☐ B 48☐ C 50☐ D 49.8

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Q4 · Rank 2 / Core

If the demand function is $P = 100 - 4Q$, find an expression for total revenue in terms of Q .

☐ A $100Q - 4Q^2$ ☐ B $100 - 4Q$ ☐ C $100Q - 4Q$ ☐ D $100 - 8Q$

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Q5 · Rank 2 / Core

If $AC = 15/Q + 2Q + 9$, write down an expression for the marginal cost function.

☐ A $2 - 15/Q^2$

☐ B $4Q + 9$

☐ C $2Q + 9$

☐ D $15 + 2Q^2 + 9Q$

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Q6 · Rank 2 / Core

The fixed costs of a good are 100 and the variable costs are $2 + Q/10$ per unit. Find an expression for marginal cost.

☐ A $2 + Q/10$

☐ B $100 + 2Q + Q^2/10$

☐ C $2 + Q/5$

☐ D $100 + Q/5$

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Q7 · Rank 3 / Application

If $P = 100 - 4Q$, estimate the change in total revenue brought about by a 0.3-unit increase in output from a current level of 12 units.

☐ A 0.84☐ B 4☐ C 1.2☐ D 15.6

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Q8 · Rank 3 / Application

Fixed costs are 100 and variable costs are $2 + Q/10$ per unit. Evaluate MC at $Q = 30$ and hence estimate the change in TC for a 2-unit increase in output from 30. Choose the estimated change.

☐ A 8☐ B 16.4☐ C 10☐ D 16

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Q9 · Rank 3 / Application

Total cost is $TC = 250 + 20Q$. Marginal revenue is 18 at $Q = 219$. If production increases slightly from 219, would you expect profit to increase, decrease or stay the same?

- ☐ A Decrease: marginal profit is -2 .
- ☐ B Increase: marginal profit is 2.
- ☐ C Stay the same: fixed cost is constant.
- ☐ D Increase: marginal profit is 18.

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Q10 · Rank 4 / Challenge · OPTIONAL

A firm has demand $P = 100 - 4\sqrt{Q} - 3Q$. Estimate the change in TR when output rises by 0.25 units from 9, and compare it with the exact change. Choose (estimate, exact change), with the exact change rounded to two decimal places.

- ☐ A (7, 7)
- ☐ B (28, 6.78)
- ☐ C (7, 6.78)
- ☐ D (6.78, 7)

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